

ZARYA PLATFORM

PROJECT GUIDE

Architecture · APIs · AI Engine · Testing · Deployment

WEBSITE <https://futureai-gamma.vercel.app>

BACKEND API <https://zarya-backend-2-production.up.railway.app/api/v1>

REPOSITORY github.com/abdullah143339-byte/zarya

DOCUMENT ZARYA_GUIDE.pdf · English

OWNER Abdullah Fuji · abdullah143339@gmail.com

About This Guide

A complete, professional reference covering every layer of ZARYA — from frontend and backend architecture to database, APIs, the multi-provider AI Router, the full testing campaign that was executed, and the live production deployment.

Version 2.0 · September 2026

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Platform Overview

ZARYA ("Think Beyond Social") is an AI-first social platform where people connect, learn, build, collaborate and grow. It combines a full social network (feed, stories, communities, real-time chat) with embedded AI assistants, a coding workspace, a learning hub, portfolios and a digital marketplace — all inside a single, premium, dark-themed application.

Purpose: An all-in-one AI-powered ecosystem: social + learning + AI + commerce

Maturity: Production build — live, tested, deployed and monitored

Audience: Everyone; scaled to students, developers, creators and professionals

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Technology Stack

The platform is split into three clean layers: a modern Next.js frontend, a modular NestJS backend, and a persistent SQLite database managed with Prisma.

Layer

Frontend	Next.js 16 · React 19	UI, SSR, routing
Styling	Tailwind CSS v4 · shadcn/ui	Premium glass UI
Motion / 3D	Framer Motion · React Three Fiber	Animations, 3D
Backend	NestJS 11 · Node.js	Modular REST API
Real-time	Socket.io · WebSocket gateway	Live chat & alerts
Database	SQLite · Prisma ORM 6	40+ tables, type-safe
Auth	JWT · Passport · 2FA	Google/GitHub OAuth
File Uploads	Multer	Images, video, media
Email	Nodemailer	Reset & verification
AI	Multi-provider AI Router	Chat, code, images
Docs	Swagger UI (/docs)	Live API reference

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Database Schema (Highlights)

SQLite is the single source of truth. On the live server the database file lives on a durable Railway volume, so it survives every redeploy and restart.

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Core: users, profiles, sessions, accounts (OAuth), security_events

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Social: posts, comments, reactions, shares, bookmarks, followers, stories, story_views

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Messaging: conversations, messages (text, media, voice), notifications

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Community & Commerce: communities, marketplace, products, orders, reviews, wallets

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Learning: courses, lessons, notes, flashcards, tasks, analytics, certificates

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AI & Ops: ai_conversations, ai_messages, audit_logs, feature_flags, settings

ORM: Prisma with automatic migration on boot (prisma db push)

Persistence: Railway volume zarya-backend-2-volume (500 MB) mounted at /app/data

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REST API Surface

The frontend talks to the backend over HTTPS. Every authenticated call sends an Authorization: Bearer <JWT> header; refresh tokens live in an HttpOnly cookie.

Base URL: <https://zarya-backend-2-production.up.railway.app/api/v1>

Response Envelope: { success: true, message, data }

Method

GET	/health	Liveness check
POST	/auth/register	Create account
POST	/auth/login	Login (identifier + password)
GET	/auth/google	Google OAuth sign-in
POST	/auth/refresh	Rotate JWT tokens
GET/POST	/users, /users/:id	Profiles & social graph
GET/POST	/posts, /posts/feed	Create & read feed
POST	/posts/:id/comment	Comments
POST	/posts/:id/react	Reactions
GET/POST	/messages/conversations	Chat threads
POST	/messages/:id/messages	Send message

POST	/uploads/post-media	Upload images/video
POST	/ai-router/chat	AI assistant (routed)
GET	/ai/conversations/:id/messages	AI chat history
GET	/learning/*	Courses & lessons
GET	/stories/*	24-hour stories
GET	/admin/*	Admin & moderation

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AI Router — Multi-Provider Engine

The AI Router inspects the user's request, picks the best provider, and automatically falls back to the next one if any provider fails. This guarantees a working AI response even during provider outages.

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Chat: Groq !' Gemini 2.5 Flash !' OpenRouter !' DeepSeek !' Mistral

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Code: generate, review, debug, explain, refactor (per language)

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Images: Flux (BFL) !' FAL AI !' Stability AI (SD3)

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Translation + automatic language detection

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Deep Search: live web search combined with an AI summary

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Surface: global assistant widget, AI conversations, portfolio generator, learning tutor

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Testing Campaign (Executed)

ZARYA was validated at every layer with live, production-grade tests. The sections below list exactly what was tested, with which tool, and where it ran.

API smoke tests — all green

Health, register, login (identifier + password), refresh and logout tested against the live Railway backend via REST calls (curl / Invoke-RestMethod).

Auth & token flow

JWT access token + HttpOnly refresh cookie verified; expired-invalid token rejected with 401; validation errors returned as structured 400 messages.

AI chat verified end-to-end

POST /ai-router/chat returned a real Gemini 2.5 Flash reply through the deployed backend (multi-provider routing functional).

Image upload + serving

POST /uploads/post-media returned the correct public HTTPS URL; GET on that URL returned 200 with correct image/jpeg content-type (static serving verified).

Post lifecycle

Created an IMAGE post via API, confirmed it appears in /posts/feed with media[0].url correctly resolved, then fetched the URL (200) and cleaned up the test post.

Headless browser E2E (Edge + Puppeteer)

D:\My Work\APP\tools\e2e — automated signup through the live website, landing on /dashboard, and the AI widget replying from the browser context.

Media URL integrity fix

Discovered uploaded URLs carried localhost host (proxy behind Railway); fixed with APP_URL + forwarded-header fallback and an idempotent boot script (fix-media-urls.cjs) that rewrites legacy URLs to the public domain.

Database persistence on redeploy

Verified an account created before a redeploy remained available afterwards — SQLite now survives restarts via the Railway volume.

CSP / env-safe deploy

Verified the frontend CSP header lists the new backend domain, and the deployed JS bundle carries the clean NEXT_PUBLIC_API_URL (a trailing "n" bug had broken all /api/v1 calls).

Super admin bootstrap

ensure-admin.cjs promotes abdullah143339@gmail.com to SUPER_ADMIN on every boot; verified via JWT role claim.

E2E harness location: D:\My Work\APP\tools\e2e (Puppeteer Core + Microsoft Edge, pdfkit, test-signup.cjs, test-full.cjs)

Tooling: REST clients (curl, Invoke-RestMethod) · Headless Chromium/Edge · Swagger /docs · Browser dev tools

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Key Defects Found & Fixed (Summary)

Issue

All API calls returned 404	NEXT_PUBLIC_API_URL had a stray trailing "n"	Re-set env cleanly with --value flag; verified bundle
Browser blocked API calls	Hardcoded old Railway domain in CSP	Updated backendDomain in next.config.ts; redeployed
Blank post images	Upload URL built with internal localhost host	APP_URL env + forwarded-host fallback + URI -fix script
Data wiped on redeploy	No persistent volume attached	Created Railway volume mounted at /app/data
Google sign-in redirect rejected	Callback enforced by old .env value	GOOGLE_CALLBACK_URL + FRONTEND_URL set on server

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Deployment (Live)

ZARYA runs as two independent cloud services, deployed through CLI pipelines and monitored with health checks.

Frontend: Vercel !' https://futureai-gamma.vercel.app (CI: vercel --prod)

Backend: Railway !' https://zarya-backend-2-production.up.railway.app (CI: railway up --detach)

Region: Railway San Francisco (sfo) · edge-cached frontend

Database: SQLite on persistent 500 MB volume (survives redeloys)

Key env groups: Provider keys (Groq, Gemini, OpenRouter, DeepSeek, Mistral, FAL, Stability) · JWT secrets · CORS_ORIGINS · FRONTEND_URL · Google OAuth creds · APP_URL · DATABASE_URL

Health check: GET /api/v1/health !' 200 (monitored after every deploy)

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Every backend boot runs: prisma db push !' cleanup scripts !' ensure-admin !' fix-media-urls !' main service.

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Sensible upgrade path: PostgreSQL + object storage (S3/Cloudinary) when scale demands; architecture already isolates these modules.

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Feature Highlights

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AI-powered feed with trending content and recommendations

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Google + GitHub one-click OAuth with optional 2FA

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Real-time private & group chat with voice notes

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AI Router: chat, code, images, translation, deep search

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Learning hub: courses, notes, flashcards, AI study assistant

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Communities, stories, posts, reactions, shares, bookmarks

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Portfolio builder with AI generation

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Marketplace with cart, orders and reviews

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Reputation system and full admin panel

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Owner & Repository

Owner: Abdullah Fuji · abdullah143339@gmail.com

Source: <https://github.com/abdullah143339-byte/zarya>

Docs PDF: ZARYA_GUIDE.pdf (this document)

For access to production secrets, provider keys or the admin panel, contact the owner directly.

All test accounts are cleaned automatically on every backend boot.

ZARYA — Think Beyond Social

Built with precision · Tested against production · Ready for the world.

